Engineering Applications of Artificial Intelligence 162 (2025) 112329

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

Research paper

A novel high-accuracy graph neural network-based rumor detection method☆

Xi Xiao [a,b,c,f], Zeming Wu [a,b,c], Chengzong Cai [a], Tian Bian [d], Guangwu Hu [e,*], Qing Li [f], Cheng Huang [g]

[a] Shenzhen International Graduate School, Tsinghua University, Shenzhen, China
[b] Key Laboratory of Data Protection and Intelligent Management (Sichuan University), Ministry of Education, Chengdu, China
[c] Key Laboratory of Cyberspace Security, Ministry of Education of China, Zhengzhou, China
[d] University of Hong Kong, Hong Kong, China
[e] Shenzhen University of Information Technology, Shenzhen, China
[f] Peng Cheng Laboratory, Shenzhen, China
[g] School of Cyber Science and Engineering, Sichuan University, Chengdu, China

ARTICLE INFO

Keywords:
Social media
Graph convolutional networks
Rumor detection
Artificial Intelligence applications
Deep learning implementation

ABSTRACT

Rumors spreading on social media platforms result in potential damages. A precise rumor detection mechanism can help form a healthy public opinion environment. In recent years, deep learning-based rumor detection methods, especially graph model-based ones, have risen and reached promising performance. However, there are several defects in existing methods, which limit models from efficiently utilizing the propagation structure. In this paper, we propose a novel rumor detection model, which has high accuracy and reaches state-of-the-art performance. First, we design a powerful comprehensive rumor feature extractor that explicitly overcomes the restriction of previous Graph Neural Networks-based models. Then, by introducing Kernel Subtree features, our model acquires the capability to learn crucial local features from important nodes. Comparative experiments performed on two real-world social media platforms demonstrate that our work reaches state-of-the-art performance, which outperforms the best baseline with 1.6% and 1.9% in accuracy respectively.

1. Introduction

Hundreds of millions of users are active on social media platforms like X and Sina Weibo every day. Among normal online activities, rumors are fabricated and spread quickly, bringing chaos to public opinions. For example, as shown in Fig. 1, after the outbreak of COVID-19, many rumors like '5G causes coronavirus', 'virus is a scam' appeared on social platforms. These rumors even led to the first global infodemic (Alam et al., 2021). In recent years, with Artificial Intelligence Generated Content(AIGC) models developing, rumor fabricators can make realistic rumors with low costs. As Fig. 2 shows, a well-generated fake picture can greatly enhance the credibility of a rumor. If left unchecked, several rumors can cause a huge economic loss. According to the report,[1] the rumor of Obama being bombed caused the Dow Jones Index to decrease by 150 points. However, faced with great quantity of data on social media platforms, it is impossible to discriminate rumors manually. Hence, automated rumor detection models are needed to protect the public opinion from being polluted (Wen et al., 2014).

The propagation process of a rumor can be modeled as a directed tree (Ma et al., 2018). And the structural information of it is proven to be crucial in discriminating veracity by Rosenfeld et al. (2020). Hence, to build a precise rumor detection model, in recent research, GNN (Sperduti and Starita, 1997), which are capable of processing graph data, are adopted to detect rumors. To the best of our knowledge, our previous work, Bi-directional Graph Convolutional Network (BiGCN) (Bian et al., 2020), the first one to apply Graph Convolutional Network (GCN) (Kipf and Welling, 2017), which is an adaptation

☆ The study was supported by the Key Laboratory of Data Protection and Intelligent Management, Ministry of Education, Sichuan University and also the Fundamental Research Funds for the Central Universities (SCU2023D008), Open Foundation of Key Laboratory of Cyberspace SecurityMinistry of Education of China and Henan Key Laboratory of Network Cryptography (No. KLCS20240304), the Key Project of the Shenzhen Municipality (Granted No. JSGG20211029095545002), and the Major Key Project of PCL (No. PCL2023A06-4).

* Corresponding author.
E-mail addresses: xiaox@sz.tsinghua.edu.cn (X. Xiao), tianbian@link.cuhk.edu.hk (T. Bian), hugw@sziit.edu.cn (G. Hu), liq@pcl.ac.cn (Q. Li), codesec@scu.edu.cn (C. Huang).

[1] https://cn.nytimes.com/usa/20130425/c25twitter/.

https://doi.org/10.1016/j.engappai.2025.112329
Received 17 March 2025; Received in revised form 1 September 2025; Accepted 12 September 2025
Available online 24 September 2025
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Fig. 1. Rumors about COVID-19 on twitter.

Fig. 2. Generated fake pictures of arrested Trump.

of the Convolutional Neural Network(CNN) to the graph domain, to detect rumors. The two subnetworks of BiGCN, i.e. Top-Down GCN (TD-GCN) and Bottom-Up GCN (BU-GCN), extract features from the rumor propagation and dispersion process respectively. In some recent works (Ma et al., 2023a; Sun et al., 2022; Xu et al., 2023), BiGCN was frequently adopted as a structural feature extractor.

However, there are two significant drawbacks in these methods:

- The first problem comes from data utilization. In previous GNN rumor detection models, the input data is formed by node texts and edges between nodes. These models first ignore the metadata (e.g., user information, account verification) and the feature of crucial nodes.
- The second problem comes from the data processing of models using BiGCN. BiGCN independently extracts rumor propagation and dispersion features. This way causes interactions between features from two directions to be discarded, which leads to coarse node representations.

To address these problems, based on the previous work (Bian et al., 2020), we extend the original method and propose a novel rumor detection model named the Attention-enhanced Graph Convolutional Network with Kernel Subtree (AttGCN-KS). To solve the first and second mentioned problems, in AttGCN-KS, we introduce the concept of Kernel Subtree (KS) (Wei et al., 2021b). KS is defined as the subtree which contains the most influential post and its children nodes. Then a set of features extracted from KS can be fed to downstream models. In the original work (Wei et al., 2021b), the authors showed that given KS features, the models can still perform well even when rumor events have a small number of nodes. A combination of KS and graph models is expected to produce a performance improvement. In combination, fusing the representations from different domains (graph structural representations and KS features) is a crucial step. Among various cross-domain feature fusing methods, Low-rank Multimodal Fusion (LMF) (Liu et al., 2018) is frequently used. It has a relatively low computational complexity and meanwhile maintains a considerable performance. Further, with regard to the second problem, we design a cross-directional attention module that can generate precise graph representations by comprehensively processing features from TD-GCN and BU-GCN. In summary, we list the main contributions of this paper as follows:

- We design an attention layer that fuses the bidirectional features of top-down propagation and bottom-up dispersion. To the best of our knowledge, it is the first attempt to model the interactions between bidirectional features.
- Based on BiGCN, we design a novel rumor detection model called the Attention-enhanced Graph Convolutional Network with Kernel Subtree (AttGCN-KS). In this model we introduce KS features to help the model learn from crucial nodes. And the aforementioned attention layer processes the features from TD-GCN and BU-GCN. Thus, we are able to obtain stronger representations of nodes and graphs.
- We conduct comprehensive experiments on two datasets. Several baselines are compared with AttGCN-KS, and the result shows that AttGCN-KS reaches the state-of-the-art performance, even in the task of early rumor detection. Further, the ablation experiment is performed to evaluate the importance of the cross-directional attention module and KS features.

The above mentioned datasets contain two binary labels: False Rumor (claims later verified as false) and True Rumor (claims later verified as true). Note that the term 'rumor' refers to unverified information circulating on social media, regardless of its eventual truth value.

The rest of this paper is organized as follows. Section 2 makes a brief summary of related work. Section 3 lists the background of our model. And the detail of our model is described in Section 4. Then in Section 5, we analyze the experimental results. Finally, the conclusion and limitations of our work are shown in Section 6.

2. Related work

In this section, previous rumor detection related works are classified into traditional machine learning based, deep learning(DL) based methods and graph structure based DL methods respectively.

2.1. Traditional machine learning based methods

In this decade, as the user population on social media platforms like Twitter and Sina Weibo grows exponentially, automatic rumor detection methods have drawn researchers' attention. A lot of traditional machine learning methods were used to detect rumors (Volkova et al., 2017; Rubin et al., 2016). Tacchini et al. presented a model based on logistic regression to classify hoaxes on Facebook (Tacchini et al., 2017). Castillo et al. (2011) utilized a J48 decision tree to assess the information credibility on Twitter and studied the effect of different kinds of rumor features. Yang et al. (2012) employed an SVM-based method to detect rumors on Sina Weibo based on several new effective features. Ma et al. (2017) used the kernel method to compute the similarities of rumor propagation trees to do classification.

Besides, feature engineering also greatly affects performance. Since a rumor can be organized in the form of a temporal sequence of texts, a considerable part of these methods applied feature engineering to extract linguistic and temporal features of rumor texts. Zheng et al. (2013) searched inquiry phrases in microblogs and clustered similar nodes for doing classification. Kwon et al. (2013) identified rumors via a new periodic time series model based on linguistic and structural features. AlRubaian et al. (2018) extracted multiple levels of features from tweet data and made an analysis system to measure information credibility on Twitter. Ma et al. (2015) built a time-sequential model of hand-crafted social context features to verify rumors.

2.2. Deep learning based methods

The limitation of traditional machine learning models leads to unsatisfactory performance. In this decade, the deep learning models have been applied to many tasks and reached considerable performance. For the rumor detection task, models like Recurrent Neural Networks (RNN) and the attention mechanism play important roles. For example, Ma et al. (2016) first applied RNN to model and classify rumor events as time sequences. Ma et al. (2023b) proposed a powerful generative adversarial learning rumor detection model based on a transformer, which is one of the most powerful attention-based networks. Besides, the combination of multiple types of network brings high performance. Guo et al. (2018) detected rumors by using an LSTM model with hierarchical social information, and applied the attention mechanism to improve accuracy. In some recent works, the attention mechanism still shows its importance. Liu and Wu (2020) applied both the attention mechanism and CNN to implement an early detection model for fake news on social media. Song et al. (2021) proposed a precise early rumor detection model using both CNN and LSTM.

Those who know the truth tends to show strong positive or negative stance in the microblog text. So, the stance information hidden in microblog text would be the indicator of rumor veracity. In order to leverage the stance information, researchers combine the stance detection task together with rumor detection task to train models. By jointly learning rumor detection and other tasks like stance detection, models would get more knowledge than the single task learning. Li et al. (2019) added stance detection sub-task to build an LSTM-based rumor detection model. Cheng et al. (2020) built a VAE-aided multi-task rumor classifier, conducting the rumor detection together with three sub-tasks. Zhang et al. (2021) designed a novel GRU-based rumor detection model including a stance classification task and text-visual multi-modal learning.

2.3. Graph structure based methods

In recent works, researchers have taken the graph structural information into consideration. Rosenfeld et al. (2020) proved the graph structure of rumor propagation is crucial in doing classification. Ma et al. (2018) extracted high-level hidden features from graph structure by using a tree-structured Recursive Neural Network (RvNN). To extract structural data in a definitive approach, Liu et al. (2024) defined a series of subgraphs in the propagation trees, and learned their encoding to assist the graph self-attention based rumor detection model. To fully exploit interactions between posts and users, Tu et al. (2021) used all propagation trees to build a weighted union graph and extracted features by CNNs. Bai et al. (2023) built a dual attention network to exploit user features and propagation tree features.

In recent works, GNN models are frequently used. They can sufficiently capture complex relationships between nodes and learn the representation of a whole graph. Rosenfeld et al. (2020) used only graph information of the rumor propagation process to classify rumors and proved the global structural features are useful in improving the performance. Bian et al. (2020) first introduced GCN to the rumor detection task and reached the state-of-the-art performance. Some of the works even added multitask-learning into training models. Wan et al. (2023) designed a multitask-learning rumor detection method, in which the detection time, accuracy and stability are all considered into training objectives. As the structural information is deeply involved in training GNN models, researchers have paid attention to utilizing it effectively. Wei et al. (2021a) proposed a GCN rumor detection model with a Bayesian edge enhancement method, which excludes uncertainty of edges and exploits latent relations between nodes. Ma et al. (2023a) utilized contrastive learning and a beta mixture model with GCN to build rumor detection models. Beyond the GCN, other GNN models were also adopted to build rumor detection methods. Yang et al. (2024) built a GNN model to extract multi-feature interactions and established an explainable rumor detection method.

3. Background

In this section, we introduce basic knowledge about GCN and Graph Attention Network(GAT) (Velickovic et al., 2018), the definition of the Kernel Subtree, and the feature fusing methods.

3.1. GCN and GAT

CNN plays an important role in deep learning methods. The ability to extract high-level representations from Euclidean data makes it a frequently-used component in the field of Computer Vision(CV). However, a great part of data like graph data is non-Euclidean, making it infeasible to apply original CNN to extract features. So, finding a generalization of CNN to the graph domain becomes a necessity. GCN provides a solution. In GCN, the convolution operation is adopted to aggregate information from a node and its neighbors. A general mathematical definition of a graph convolutional layer is as follows:

$$\mathbf{Z}^{l+1} = M(\mathbf{Z}^l, \mathbf{A}). \tag{1}$$

In the above formula, $\mathbf{Z}^{l+1}$ is the $(l+1)$th layer output embedding. Specifically, $\mathbf{Z}^0 = \mathbf{X}$, which denotes the input features. $\mathbf{A}$ is the adjacency matrix, describing adjacency relationships between nodes. M denotes the message propagation function, which computes $\mathbf{Z}^{l+1}$ with $\mathbf{Z}^l$ and $\mathbf{A}$ as parameters. Obviously, the selection of the message propagation function M plays an extremely important role. The one in the first-order approximation of Chebnet (1stChebNet) (Kipf and Welling, 2017; Defferrard et al., 2016) is frequently used among various functions (Bruna et al., 2014). Its mathematical definition is as below:

$$\mathbf{Z}^{l+1} = \sigma(\mathbf{D}^{-\frac{1}{2}}\tilde{\mathbf{A}}\mathbf{D}^{-\frac{1}{2}}\mathbf{Z}^l\mathbf{W}^{l+1}). \tag{2}$$

In the above formula, $\tilde{\mathbf{A}} = \mathbf{A}+\mathbf{I}$ is the adjacency matrix with self-loop and $\mathbf{D}$ represents a diagonal matrix in which $\mathbf{D}_{ii} = \sum_j \tilde{\mathbf{A}}_{ij}$. And σ is an activation function, e.g., the ReLU or sigmoid function. $\mathbf{W}^{l+1}$ denotes the learnable weight parameters in the $(l+1)^{th}$ layer and the main task of a GCN model is to learn a proper set of $\mathbf{W}^{\mathbf{l}}$.

Similarly, GAT (Velickovic et al., 2018) is an adaptation of the attention network to the graph domain. It leverages attention mechanisms to learn the importance of different nodes in a graph while aggregating information from their neighbors. Similar to GCN, GAT has been widely used in various tasks, such as node classification, link prediction, and graph classification. GAT calculates attention coefficients for each pair of nodes, indicating how much attention should be given to each neighbor when aggregating information. Given a graph with n nodes, we represent it as a set of node features X_i where i ranges from 1 to n. The formula for computing attention coefficients in GAT is as follows:

$$e_{ij} = \text{LeakyReLU}(\vec{a}^T[\mathbf{W}X_i \oplus \mathbf{W}X_j]). \tag{3}$$

In this formula, e_{ij} represents the attention coefficient between node i and its neighbor j. The concatenation operator "$\oplus$" combines the transformed feature vectors $\mathbf{W}X_i$ and $\mathbf{W}X_j$. The weight vector $\vec{a}$ and matrix $\mathbf{W}$ are learnable parameters. LeakyReLU is a non-linear activation function that introduces a small negative slope for negative values to prevent dead neurons. It is used to add non-linearity to the attention mechanism.

After obtaining the attention coefficients, we can compute the attention weight vector $\vec{\alpha}_i$ by applying a softmax function:

$$\vec{\alpha}_i = \text{softmax}(\vec{e}_i). \tag{4}$$

$\vec{e}_i$ denotes the attention coefficient vector formed by e_{ij}. The softmax function ensures that the attention weight vector $\vec{\alpha}_i$ sums up to 1 across the neighbors of node i. Finally, we aggregate the features of the neighbors by taking a weighted sum using the attention weights:

$$H_i = \sum_j \alpha_{ij}\mathbf{W}X_j. \tag{5}$$

In this equation, H_i represents the updated representation of node i. This aggregation step considers the importance of each neighbor according to the attention weights.

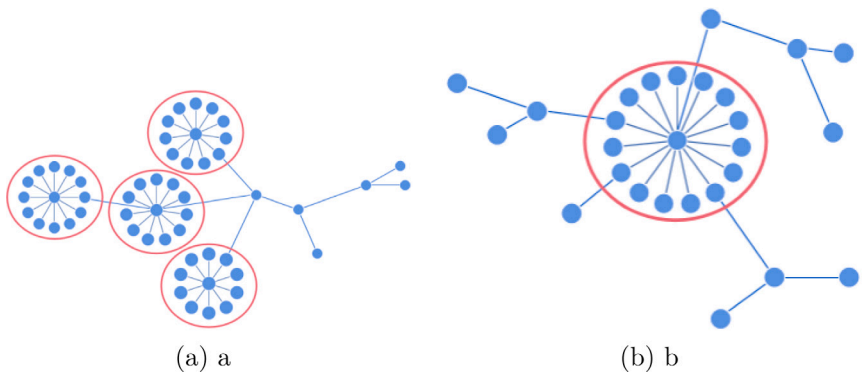

Fig. 3. (a) A non-rumor propagation tree, (b) A rumor propagation tree.

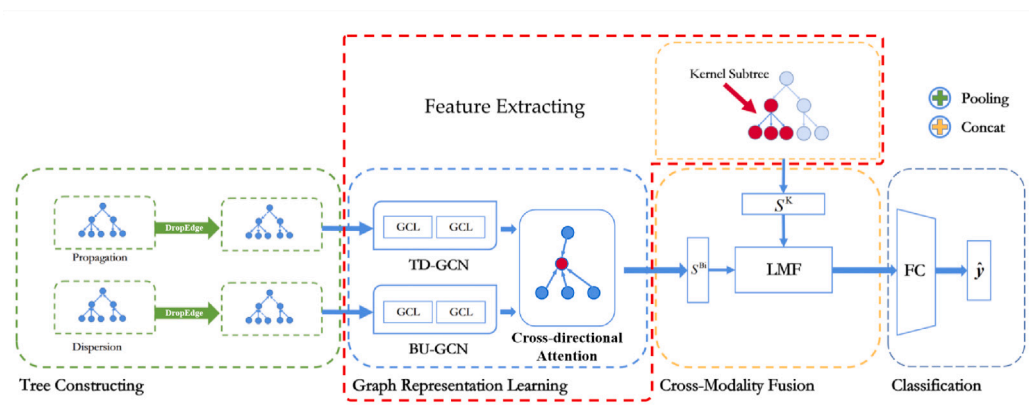

Fig. 4. Architecture of AttGCN-KS: The tree constructing module builds the propagation and dispersion trees. In the Feature Extracting module, representations of the graph structure and KS are obtained via the graph representation Learning sub-module and the extraction of KS features respectively. Then two kinds of representation are fused in the Cross-modality fusion module. Finally, the predicted label is generated by the last module.

3.2. Kernel subtree

Fig. 3 shows the propagation trees of rumor and non-rumor. Nodes represent the posts. The red circles mark some influential posts. These posts have much more reposts and replies, which indicates that these account owners are much more influential than common ones, e.g., celebrities or official media. It can be observed that in a non-rumor propagation tree, there are several influential posts, meanwhile, much fewer influential posts exist in a rumor propagation tree. It is perhaps that influential accounts are willing to repost the verified information and unlikely to repost unverified or wrong information. Besides, the difference between the characteristics of influential posts in the two kinds of trees can also be helpful in distinguishing rumors.

In order to obtain important information from these influential posts, Wei et al. defined the Kernel Subtree (KS) as a tree that contains the node with the largest degree and its child nodes (Wei et al., 2021b). Many informative features can be extracted in the kernel subtree, and they can be classified into three kinds. In the original paper (Wei et al., 2021b), there are 31 types of KS features. But while running the experiment, more features were included, the details can be found in Section 5.

3.3. Feature fusing methods

The high-level structural features extracted by GCN is so different from the kernel subtree feature that we can regard them as different modalities like visual and audio modalities. In fact, the information that they contain is complementary. So to fully mine their complement parts to each other, a proper cross-domain feature fusion method becomes a necessity. A popular method (Zadeh et al., 2017) is to build a matrix by operating the outer product between those vectors from different modalities and process that matrix by neural networks. This method has a significant drawback: the number of parameters explodes as the dimension of vectors grows, bringing a slow and big model. To avoid this problem, Low-rank Multimodal Fusion (LMF) (Liu et al., 2018) was proposed. The mathematical description of LMF is given below:

$$H = \bigwedge_{m=1}^{M} \left[\sum_{i=1}^{r} \mathbf{W}_m^i \cdot Z_m \right]. \tag{6}$$

In the above formula, M denotes the number of modalities, Z_m is the feature vector of the mth modality. Λ is the Hadamard product. A key parameter of LMF is the rank r. By adjusting r to a proper number, we can greatly reduce the complexity of a fusion model without sacrificing the performance. In our work, we utilize LMF to fuse features from the previous attention module and the kernel subtree.

4. AttGCN-KS model architecture

In this section, we present a panoramic view of the architecture of AttGCN-KS and describe the details of each part. For the convenience of understanding our method, Table 1 describes the used mathematical symbols.

Fig. 4 shows the architecture of our model. There are four modules in our model: Tree Constructing, Feature Extracting, Cross-modality Fusion, and Classification. In a nutshell, in the first module, we build the propagation and dispersion trees of a rumor event. Then, we extract

Table 1
Notation.

Symbol	Dimension	Description
$\mathbf{A}$	$n \times n$	Adjacency matrix of a propagation tree
$\hat{\mathbf{A}}$	$n \times n$	Normalized adjacency matrix
$\mathbf{A}^{TD}/\mathbf{A}^{BU}$	$n \times n$	Adjacency matrix in TD-GCN/BU-GCN
$\mathbf{X}$	$n \times d$	Raw feature matrix
$\vec{V}_{root}$	d	Root feature vector in $\mathbf{X}$
$\mathbf{X}^{root}$	$n \times d$	n copies of the root feature vector $\vec{V}_{root}$
$\mathbf{W}_i^{TD}/\mathbf{W}_i^{BU}$	$v_{i-1} \times v_i$	Weight matrix in the ith GCN Layer of TD/BU-GCN
$\mathbf{H}_i^{TD}/\mathbf{H}_i^{BU}$	$n \times v_i$	Hidden representation matrix in the ith GCL of TD/BU-GCN
$\mathbf{H}_{root}^{TD}/\mathbf{H}_{root}^{BU}$	$n \times v_i$	n copies of root representation in the 1st GCL
$\tilde{\mathbf{H}}_i^{TD}/\tilde{\mathbf{H}}_i^{BU}$	$n \times (v_{i-1} + v_i)$	Representation matrix after root feature enhancement
$\vec{a}^{TD}/\vec{a}^{BU}/\vec{a}^{Bi}$	$2v_{att}$	Vectors of computing attention coefficients
$\mathbf{W}_{att}^{TD}/\mathbf{W}_{att}^{BU}/\mathbf{W}_{att}^{Bi}$	$v_2 \times v_{att}$	Weight matrices of computing attention coefficients
S^F	v_{att}	Final output vector of CAL
S^K	v_K	Feature vector of the kernel subtree
$\mathbf{W}_F^{Bi}$	$r \times v_h \times (v_{att})$	Weight matrix for BiGCN output vector in LMF
$\mathbf{W}_F^K$	$r \times v_h \times v_K$	Weight matrix for kernel subtree feature vector in LMF
h	v_h	The fusion vector

KS features and obtain high-level structural features by feeding two trees into a bi-directional graph convolutional network. After that, in the third module, the two kinds of features are fused with each other. Finally, a fully-connected and a softmax layer outputs the predicted label.

4.1. Tree constructing

In our method, the input data is a batch of events. Every event is organized in the form of a propagation tree. A propagation tree contains two parts. The first one is the features of nodes, which are defined as Term Frequency-Inverse Document Frequency(TF-IDF) vectors extracted from node texts. The second is the adjacency matrix. We treat the propagation tree as a directional graph and calculate its adjacency matrix $\mathbf{A}^{TD}$, in which an edge represents a repost or comment relation between two nodes. But a propagation tree only represents a top-down deep rumor propagating process from the source node, in which the influences children nodes act on the parent node is ignored. Besides deep propagation along a relationship chain, the wide dispersion across a social community is another important characteristic of rumor. Hence, to retain the child-to-parent influences and model the wide dispersion, we construct another tree, the dispersion tree. The dispersion tree shares the same node features with the propagation tree. However, the adjacency matrix of a dispersion tree is transposed to the one of a propagation tree, i.e.,

$$\mathbf{A}^{BU} = (\mathbf{A}^{TD})^T. \tag{7}$$

The reversed edges would help GCN generate a representation of parent nodes using information from children nodes.

To alleviate the overfitting problem, DropEdge (Rong et al., 2020) is applied on trees to randomly disconnect edges with the probability of p (i.e., the drop rate), forcing the model to learn from the global structure of rumor events. The adjacency matrix $\mathbf{A}$ is modified by DropEdge following the formula:

$$\mathbf{A} \leftarrow \mathbf{A} - \mathbf{A}^{Drop}, \tag{8}$$

where $\mathbf{A}^{Drop}$ is generated by randomly sampling the original edges with the probability of p.

4.2. Feature extracting

To process the propagation and dispersion trees, the Top-Down GCN (TD-GCN) and Bottom-Up GCN (BU-GCN) are created. TD-GCN and BU-GCN share the same structure and hyperparameters but different weight matrices. The propagation tree and dispersion tree are specified as the input to TD-GCN and BU-GCN respectively. The outputs of two GCNs can be regarded as top-down and bottom-up high-level representations of input events. After the step of tree construction, the high-level hidden representations of propagation and dispersion can be obtained by TD-GCN and BU-GCN respectively. In the following, we explain how TD-GCN works in detail.

Denote the number of nodes as n, and the original feature matrix of each node as $\mathbf{X} \in \Re^{n \times d}$ and the ReLU function as $\sigma(\cdot)$, TD-GCN learns hidden representations of propagation as below:

$$\mathbf{H}_1{}^{TD} = \sigma(\hat{\mathbf{A}}^{TD}\mathbf{X}\mathbf{W}_1{}^{TD}), \tag{9}$$

$$\tilde{\mathbf{H}}_1^{TD} = \mathbf{H}_1{}^{TD} \oplus \mathbf{X}^{root}, \tag{10}$$

$$\mathbf{H}_2{}^{TD} = \sigma(\hat{\mathbf{A}}^{TD}\tilde{\mathbf{H}}_1^{TD}\mathbf{W}_2{}^{TD}), \tag{11}$$

$$\tilde{\mathbf{H}}_2^{TD} = \mathbf{H}_2{}^{TD} \oplus \mathbf{H}_1{}^{root}. \tag{12}$$

The root features are extremely crucial to discriminate a rumor. Hence, in order to emphasize more on root features, the root feature vector $\vec{V}_{root} \in \Re^d$ in $\mathbf{X}$ is copied n times to form $\mathbf{X}^{root}$, and concatenated (denoted as $\oplus$) with $\mathbf{H}_1^{TD}$. Next, the root feature vector in $\tilde{\mathbf{H}}_1^{TD}$ is copied n times to form $\mathbf{H}_1^{root}$, and concatenated with $\mathbf{H}_2{}^{TD}$to get enhanced hidden representations $\tilde{\mathbf{H}}_2^{TD}$. BU-GCN has the same structure as TD-GCN. The difference between TD-GCN and BU-GCN is the adjacency matrix, so here we skip the introduction of BU-GCN. In our previous work (Bian et al., 2020), TD-GCN and BU-GCN are combined into a Bi-directional Graph Convolutional Network (BiGCN).

However, to further model the interactions between nodes, instead of simply concatenating operation (Bian et al., 2020) in the original work, we design a modified attention module called the Cross-directional Attention Layer (CAL) to learn the final representation S^F. For node i, we first compute the attention coefficients. The coefficients are formed by three parts which stand for the father node, children nodes and node i itself. For the father node f, its influences on node i are modeled in TD-GCN, so we use their top-down hidden representations to compute the coefficient:

$$e_{fi}^{TD} = \text{LeakyReLU}(\vec{a}^{TD}[\mathbf{W}_{att}^{TD}H_f^{TD} \oplus \mathbf{W}_{att}^{TD}H_i^{TD}]), \tag{13}$$

where H_f^{TD} and H_i^{TD} denote the hidden representation vectors from TD-GCN. Similarly, the coefficients for children nodes are obtained by processing their bottom-up hidden representations.

$$e_{ij}^{BU} = \text{LeakyReLU}(\vec{a}^{BU}[\mathbf{W}_{att}^{BU}H_i^{BU} \oplus \mathbf{W}_{att}^{BU}H_j^{BU}]), \tag{14}$$

where node j belongs to the children nodes of i. To compute the attention coefficient for node i itself, hidden representations from both TD-GCN and BU-GCN are also needed:

$$e_i^{Bi} = \text{LeakyReLU}(\vec{a}^{Bi}[\mathbf{W}_{att}^{BU}H_i^{BU} \oplus \mathbf{W}_{att}^{TD}H_i^{TD}]). \tag{15}$$

We employ LeakyReLU to preserve gradient flow for negative attention coefficients, which is crucial when modeling conflicting propagation patterns in rumor scenarios. Unlike existing attention mechanisms that operate within single directions, our CAL explicitly models three-way interactions: parent influence (TD), children influence (BU), and self-representation (Bi). This captures both supportive and contradictory response dynamics simultaneously, addressing BiGCN's limitation of processing propagation directions independently.

After obtaining coefficients, we use the softmax function to compute the attention weights:

$$\alpha_{fi}^{TD} = \frac{exp(e_{fi}^{TD})}{\sum_{k\in c(i)} exp(e_{ik}^{BU}) + exp(e_{fi}^{TD}) + exp(e_i^{Bi})}, \tag{16}$$

$$\alpha_{ij}^{BU} = \frac{exp(e_{ij}^{BU})}{\sum_{k\in c(i)} exp(e_{ik}^{BU}) + exp(e_{fi}^{TD}) + exp(e_i^{Bi})}, \tag{17}$$

$$\alpha_i^{Bi} = \frac{exp(e_i^{Bi})}{\sum_{k\in c(i)} exp(e_{ik}^{BU}) + exp(e_{fi}^{TD}) + exp(e_i^{Bi})}, \tag{18}$$

In the above formula, $c(i)$ represents the set of children nodes. The attention weights α_{fi}^{TD}, α_{ij}^{BU} and α_i^{Bi} represent the importance of influences from the upper-stream, down-stream and the node itself. By aggregating features from adjacent nodes, we obtain the final hidden representations for node i and the whole graph:

$$\begin{aligned} H_i^F = &\sum_{k\in c(i)} \alpha_{ik}^{BU} \mathbf{W}^{BU} H_k^{BU} + \alpha_{fi}^{TD} \mathbf{W}^{TD} H_f^{TD} \\ &+ \alpha_i^{Bi} \frac{(\mathbf{W}^{BU} H_i^{BU} + \mathbf{W}^{TD} H_i^{TD})}{2} \end{aligned} \tag{19}$$

$$S^F = AVG(\mathbf{H}^F) \tag{20}$$

where $AVG(\cdot)$ is the average pooling operation and $\mathbf{H}^F$ is the matrix formed by high-level node representations H_i^F. By applying this module, instead of modeling the propagation and dispersion separately, we process the interactions between TD-GCN and BU-GCN features, bringing more precise node and global representations.

By contrast, the extraction of KS features is much simpler. We extract KS features as Section 3.2 describes, and transform them into a vector. Then, through a two-layer fully connected network, the KS representation S^k is obtained.

4.3. Cross-modality fusion

From the above processes, we obtain the high-level structural representation S^F and KS feature vector S^K. The two kinds of features can be regarded as the learning-based feature and the knowledge-based feature respectively. The difference between the two types of features is large enough to treat them as they come from different modalities. Hence, in this stage, fusing two features into one vector is operated before doing classification.

Specifically, we adopt Low-rank Multimodal Fusion (LMF) (Liu et al., 2018) to fuse S^F and S^K. LMF is an optimized version of the frequently used outer production method (Zadeh et al., 2017) in fusing features from different modalities, it greatly reduces the number of parameters without sacrificing the performance. The following formula describes the fusing process to get the fusion vector H:

$$H = (\sum_{i=1}^{r} \mathbf{W}_i^{Bi} \cdot S^F) \circ (\sum_{j=1}^{r} \mathbf{W}_j^{K} \cdot S^K). \tag{21}$$

Here, $\mathbf{W}_i^{Bi} \in \Re^{v_h \times 2(v_1+v_2)}$ and $\mathbf{W}_i^{K} \in \Re^{v_h \times v_K}$ are the fusion parameters of S^F and S^Krespectively. $\circ$ represents the Hadamard product, i.e., the element-wise product between tensors.

4.4. Classification

Finally, the predicted label $\hat{y}$ is achieved by putting the fusion vector H into a fully connected and softmax layer as below:

$$\hat{y} = \mathrm{softmax}(\mathrm{FC}(H)). \tag{22}$$

$\hat{y}$ is a number among [0,1], a larger value indicates a higher probability of being a rumor. Besides, we train the whole model by minimizing the cross-entropy loss between the predicted label $\hat{y}$ and the ground-truth label y over given datasets. The formula of the loss function is described below, where $\|w\|_2^2$ denotes the L_2 regularization of all model parameters and λ is a coefficient:

$$L(y, \hat{y}) = -y\log(\hat{y}) - (1-y)\log(1-\hat{y}) + \lambda\|w\|_2^2. \tag{23}$$

Table 2
Statistics of datasets.

Statistic	Weibo2016	PHEME
# of posts	3,805,656	1,032,125
# of users	2,746,818	67,509
# of events	4664	5802
# of true rumors	2313	3830
# of false rumors	2351	1972
Avg. time length/event	2460.7 h	5.3 h
Avg. # posts/event	816	17.8
Max # posts/event	59,318	346
Min # posts/event	10	1

5. Experiments

Based on two datasets, thorough experiments on AttGCN-KS and several classical or state-of-the-art baselines are conducted. Further, we compare the early rumor detection performances of our method and the baseline models. Finally, we design ablation experiments to investigate the roles that KS and CAL play in improving performance.

5.1. Datasets

Weibo and Twitter are the most popular social media platforms in China and the U.S. respectively. Thus, to verify the ability to detect real-world rumors, we do the experiments on one real-world dataset based on Weibo: Weibo (Ma et al., 2016) (To distinguish from the website of Sina Weibo, Weibo (DALEY and KENDALL, 1965) is called Weibo2016 in the remaining parts), and one dataset based on Twitter: PHEME (Zubiaga et al., 2016). These two datasets contain two binary labels: False Rumor (F) and True Rumor (T). The events in Weibo2016 are labeled according to the Sina Community Management Center. And the label of each event in PHEME is annotated by a team of journalists (Zubiaga et al., 2016). The statistics of the two datasets are shown in Table 2.

Raw social media data undergoes standard preprocessing to extract propagation relationships. For Weibo, we parse post content from repost/reply chains. For PHEME, we extract similar relationships from source tweets and reaction. This generates structured propagation sequences with TF-IDF text features and temporal metadata.

5.2. Settings

We compare the proposed method with some classical or state-of-the-art baselines, including:

•DTC (Castillo et al., 2011): A decision tree-based rumor detection model, using hand-crafted features to do classification.

•SVM-RBF (Yang et al., 2012): A rumor classifier based on SVM and RBF kernel. Overall statistics of events are needed for it to obtain information credibility.

•SVM-TS (Ma et al., 2016): A linear SVM classifier that leverages hand-crafted features to construct a time-series model.

•RvNN (Ma et al., 2018): A rumor detection model based on tree-structured recursive neural networks with GRU units. It can learn rumor structural features via the propagation tree.

•PPC_RNN+CNN (Liu and Wu, 2018): A rumor detection approach combining RNN and CNN. It extracts high-level rumor representations by dealing with user characteristics in the rumor propagation process.

•StA-HiTPLAN(Structure Aware Hierarchical Token and Post-Level Attention Network) (Khoo et al., 2020): A novel rumor detection model based on a self-attention mechanism. Temporal information and interaction information between posts are also extracted for classification.

•BiGCN (Bian et al., 2020): Our previous work, a GCN-based rumor detection model utilizing the Bi-directional propagation structure.

Table 3
Hyperparameter settings.

Optimizer	DropEdge rate	LMF rank	Learning rate	Batch size
Adam	0.1	3	0.0005	64

•CRFB(Contrastive Learning for Rumor Detection via Fitting Beta mixture model) (Ma et al., 2023a): A contrastive learning rumor detection model, in which CNN and GCN are both applied to capture features from the propagation structure.

•DAN-Tree(Dual Attention Network on propagation Tree structures) (Bai et al., 2023): A novel rumor detection model which utilizes transformer encoding blocks as feature extractors to model the implicit relationships among posts on the propagation path.

DTC and SVM-based models are implemented with Scikit-learn[2]; PPC_R NN+CNN with Keras[3]; the rest of the methods with Pytorch.[4] All datasets are randomly split into five parts to conduct 5-fold cross-validations to obtain test results. The main hyperparameter settings in the training process are shown in Table 3. The early stopping is also performed when the validation loss stops decreasing for 10 epochs.

In the original paper that proposed KS [15], there are 31 terms of manually predefined features. These features are mostly statistical figures and can be divided into 3 classes.

- Propagation-based Features: Propagation-based features consist of several statistical features of the propagation tree, e.g., the ratio of leaf nodes in the kernel subtree to all nodes in the propagation tree.
- User-based Features: This kind of feature mainly contains user credit information, such as the follower number of the most influential node.
- Content-based Features: Content-based features are about the emotion and the stance the most influential node expresses, which may indicate the veracity of rumors, such as the percentage of some strongly emotional words in posts of the kernel subtree.

To make our model learn more information from KS, we add several other features. As Table 4 shows, the newly-added features are mainly propagation-based and user-based. There are 3 common propagation-based features for all 3 datasets. Among them, R0 is a frequently used index in epidemiology, which is defined here as the ratio of leaf nodes to non-leaf nodes. To help identify bot accounts, statistics about the repost and comment counts are introduced. A bot account usually does not have a large number of comment or repost counts. Besides, the percentage of reposts with no word in KS is added to help discriminate a certain type of bot account, which is created to falsely increase the repost count of the original post.

5.3. Comparison of results

Table 5 shows the performance of the proposed method and all the compared baselines on PHEME, Weibo2016, respectively. In these tables, for all datasets, Accuracy (Acc.) is computed over the two categories, and Precision (Prec.), Recall (Rec.), F1 measure (F1) on each category. The row of F refers to these indicators except Acc. is computed by regarding the false rumors as positive ones and that of T by regarding the true rumors as positive ones. It is worth noting that we do not test StA-HiTPLAN on the Weibo2016 dataset, because there is a matrix in StA-HiTPLAN that describes the relationship between each pair of nodes in an event. Denoting the number of nodes as n, the space complexity of that matrix is $O(n^2)$. Hence the events with hundreds of thousands of nodes in Weibo2016 make our server unable to store data while running StA-HiTPLAN.

First, from the experimental results, the deep learning models (PPC_RNN+CNN, RvNN, BiGCN, StA-HitPLAN, CRFB, DAN-Tree, AttGCN-KS) are superior to those traditional machine-learning-based methods (DTC, SVM-RBF, SVM-TS). The result is reasonable since the auto-extracted features can better represent valid information in rumor events than hand-crafted ones. This indicates the necessity of applying deep learning methods for rumor classification.

Second, our model performs better than the PPC_ RNN +CNN method on all two datasets. There are two reasons why the performance of PPC_RNN+CNN is worse. The most crucial one is both RNN and CNN ignore structural information in the rumor propagation trees. In addition, PPC_RNN+CN N only focuses on the characteristics of users. As shown in Table 2, the PHEME dataset has fewer users than the Weibo2016 dataset. It can be drawn a conclusion that it is important to introduce structural information to rumor detection models.

Compared with BiGCN and the more recent state-of-the-art methods, DAN-Tree and CRFB, AttGCN-KS still keeps its superiority. The events in Weibo2016 have a complex graph structure, which is suitable for BiGCN and CRFB to exploit structural information. DAN-Tree has lower expressive power for graph structures due to the lack of GNN models. Hence on Weibo2016, the gap between BiGCN and CRFB and AttGCN-KS becomes narrow while the one between AttGCN-KS and DAN-Tree goes larger. But in PHEME, events with much fewer nodes contain less structural information, making two GCN models unable to learn efficient high-level graph structural representation. Moreover, both of CRFB and BiGCN use TF-IDF features to represent the textual contents of nodes, which are unable to encode complex content information. Hence, the extra content information provided by KS features brings a large margin of performance between AttGCN-KS and them. In contrast, DAN-Tree uses BERT (Devlin et al., 2019) to embed texts and obtains a small performance gap between itself and AttGCN-KS. For StA-HiTPLAN, in PHEME, its time-delay embedding module can precisely map short time delays between reposts and the source post to vector representations, making its performance close to AttGCN-KS. However, the longer average time and the simpler graph limit the time-delay embedding and structural aware self-attention of StA-HiTPLAN. In conclusion, on all two datasets, AttGCN-KS reaches the highest performance.

5.4. Impact of the kernel subtree

The kernel subtree features make a great contribution to the performance of our model. So for building a more powerful model, it is necessary to investigate how they affect the performance. As mentioned in Section 3, the kernel subtree features consist of three types. To measure the impact of each type, we run a comparison experiment on two datasets. We remove a certain type of kernel subtree features and preserve the other two types to obtain three variants. Table 6 compares the performance of AttGCN-KS and these variants. In the table, No Propagation-based refers to the model in which we remove propagation-based kernel subtree features and preserve the other two types of features. No User-based and No Content-based are similar.

Obviously, the results show the extreme importance of user-based features. On Weibo2016, without user-based features, the accuracy declines with the greatest magnitude compared to the original model AttGCN-KS. And the lack of content-based features only causes an accuracy decline of 0.006. The user-based features also keep their importance on the PHEME dataset, which causes an accuracy change of 0.013. A probable explanation for this phenomenon is that the Graph Convolutional Layers (GCLs) in AttGCN-KS can effectively exploit the graph structural information, making propagation-based features less important. But GCLs only use simple TF-IDF features to represent textual content information, making them less capable of understanding complex microblog texts. Hence content-based features are more helpful than propagation-based ones. Similarly, because of the lack of user-related information in GCLs' input, user-based features in KS strongly supplement the input and improve the performance.

[2] https://scikit-learn.org.
[3] https://keras.io/.
[4] https://pytorch.org/.

Table 4
New KS features for each dataset.

Type	Weibo2016	PHEME
Propagation-based	{Timespan of propagation tree} {# of non-leaf nodes in propagation tree/# of tweets in total} {# of leaf nodes in propagation tree/# of non-leaf nodes in propagation tree (R0)}	{Timespan of propagation tree} {# of non-leaf nodes in propagation tree/# of tweets in total} {# of leaf nodes in propagation tree/# of non-leaf nodes in propagation tree (R0)}
User-based	{Verified type of the most influential node} {# of the repost and comment count of the most influential node} {Average # of the repost and comment count of users of kernel subtree} {# gender ratio of users in kernel subtree}	{# of the repost and comment count of the most influential node} {Average # of the repost and comment count of users of kernel subtree}
Content-based	{% of reposts with no words in kernel subtree}	

Table 5
Experimental results.

Method	Class	PHEME				Weibo2016			
		Acc.	Prec.	Rec.	F1.	Acc.	Prec.	Rec.	F1.
DTC	T	0.649	0.673	0.923	0.778	0.831	0.847	0.815	0.831
	F		0.417	0.118	0.183		0.815	0.824	0.819
SVM-RB	T	0.662	0.666	0.980	0.793	0.879	0.777	0.656	0.708
	F		0.528	0.045	0.084		0.579	0.708	0.615
SVM-TS	T	0.661	0.665	0.986	0.795	0.885	0.950	0.932	0.938
	F		0.375	0.016	0.031		0.124	0.047	0.059
PRC_RNN+CNN	T	0.655	0.670	0.940	0.782	0.916	0.884	0.957	0.919
	F		0.467	0.102	0.167		0.955	0.876	0.913
RvNN	T	0.816	0.832	0.889	0.860	0.908	0.912	0.897	0.905
	F		0.783	0.690	0.734		0.904	0.918	0.911
StA-HiTPLAN	T	0.854	0.891	0.887	0.889	/	/	/	/
	F		0.781	0.787	0.784		/	/	/
BiGCN	T	0.822	0.843	0.896	0.869	0.961	0.961	0.964	0.961
	F		0.774	0.676	0.721		0.968	0.965	0.967
CRFB	T	0.838	0.870	0.887	0.878	0.967	0.965	0.969	0.967
	F		0.772	0.742	0.757		0.968	0.965	0.960
DAN-Tree	T	0.845	0.888	0.872	0.879	0.958	0.946	0.972	0.958
	F		0.760	0.786	0.773		0.972	0.945	0.958
AttGCN-KS	T	**0.861**	**0.893**	0.927	**0.909**	**0.975**	**0.984**	**0.969**	**0.975**
	F		**0.816**	**0.779**	**0.797**		**0.971**	**0.980**	**0.975**

Table 6
Comparison with different KS features.

Dataset	Feature type	Acc.
Weibo2016	No propagation-based	0.973
	No user-based	0.966
	No content-based	0.971
	AttGCN-KS	0.986
PHEME	No propagation-based	0.860
	No user-based	0.849
	No content-based	0.852
	AttGCN-KS	0.862

5.5. Effect of cross-directional attention layer

In extracting precise node-level representations, CAL comprehensively combines the influences from the upper-stream and down-stream nodes. To further quantize the effect of CAL, we design an ablation experiment. In the experiment, we remove CAL in the original model and generate the global graph representation by simply concatenating and averaging TD-GCN and BU-GCN features as the previous work (Bian et al., 2020):

$$\mathbf{H}^F = \mathbf{H}^{TD} \oplus \mathbf{H}^{BU}, \tag{24}$$

$$S^{Bi} = AVG(\mathbf{H}^F). \tag{25}$$

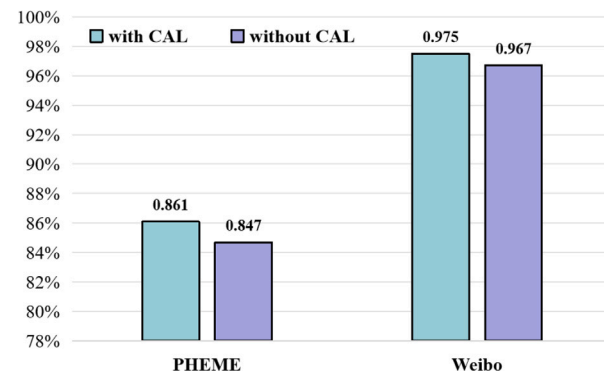

Fig. 5. Accuracy comparison between AttGCN-KS and the model without CAL.

The accuracy result is shown in Fig. 5. There exists a considerable performance margin between the two models. The reason is that there exist hidden connections between Top-down and Bottom-up features, which is informative for classification. The simple concatenation and averaging operations are unable to model these connections, while CAL can comprehensively process the relation between two kinds of features and generate more powerful representations.

6. Conclusion

In this paper, we propose a novel rumor detection model called AttGCN-KS, which utilizes KS features and a cross-directional attention module to enhance the previous BiGCN model. Its GCN part effectively extracts high-level features of the graph structural information in rumor events. The cross-directional attention module comprehensively computes global graph representations by combining features from upper-stream and down-stream. Then three kinds of features from the kernel subtree make the model focus on the most influential node and its children, which is crucial in judging the veracity. Finally, the model fuses KS features and the graph representations to discriminate rumors. Experimental result shows that our model outperforms the state-of-the-art methods on all datasets. In the future, our experimental results may spur research in various aspects including critical features and more comprehensive modules.

CRediT authorship contribution statement

Xi Xiao: Conceptualization. **Zeming Wu:** Data curation. **Chengzong Cai:** Formal analysis. **Tian Bian:** Investigation. **Guangwu Hu:** Writing – review & editing. **Qing Li:** Methodology. **Cheng Huang:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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